CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

April/May 2012

EC203 Solid State Electronics

Date: 03.05.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections and Use of scientific calculator is allowed.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required

SECTION - I

Q - 1 (a) Describe following terms briefly **[03]**
 (i) Thermistor
 (ii) Sensitor
 (iii) Photoconductor

(b) What is mobility? Derive equation for current density J and conductivity σ. **[04]**

Q - 2 (a) Prove that Fermi level lies in center of forbidden energy band in intrinsic semiconductor. **[07]**

(b) Why Silicon is prefer over Germanium in IC fabrication. Also mention advantages of Ge over Si. **[03]**

(c) The amplifier of figure 1 utilizes an n-channel FET for which $V_P = -2.0$ V and $I_{DSS} = 1.65$ mA. It is desired to bias the circuit at $I_D = 0.8$ mA, using $V_{DD} = 24$ V. Assume $r_d >> R_d$. Find V_{GS}, g_m *and* R_S **[04]**

OR

(c) The Fixed bias configuration of figure 2 had an operating point defined by $V_{GS_Q} = -2$ V and $I_{D_Q=5.625}$ mA with $I_{DSS} = 10$ mA and $V_P = -8$ *V*. For $y_{OS} = 40$ μs find out g_m, r_d, z_i, z_o and A_V. **[04]**

Q - 3 (a) Classify amplifiers on the bases of operation method (Q point) and also mention application of amplifier. **[07]**

(b) Draw and Explain V-I characteristic of idle p-n junction diode. **[03]**

(c) Discuss RC coupled multistage amplifier. **[04]**

OR

(c) Discuss Reverse Saturation Current, Rise Time, Tilt and square wave testing . **[04]**

SECTION – II

Q - 4 (a) Define following terms **[03]**
 (i) Cutin voltage in diode
 (ii) Avalanche multiplication
 (iii)Photovoltaic potential

(b) Explain Full wave rectifier using diode **[04]**

Q - 5 (a) Find the transistor currents in the figure 3. A silicon transistor with $\beta = 100$ and $I_{CO} = 20$ nA is under consideration. Also Repeat same if 2 kΩ emitter resistance is provided (figure 4). **[07]**

OR

Q - 5 (a) Consider the transistor common emitter amplifier and the h parameters are given in table 1. If $R_L = 10K$ and $R_S = 1K$, find the various gains and input and output impedance. [07]

(b) Explain Early effect/ Base width modulation. [04]

(c) Explain Thermal Runaway. [03]

Q - 6 (a) Explain the "h Parameter" analysis of a transistor amplifier. [07]

OR

(a) Explain the "hybrid - π" model for conductance and resistance. [07]

(b) Compare CE and CC configuration. [04]

(c) Explain Ebers-moll model for p-n-p transistor. [03]

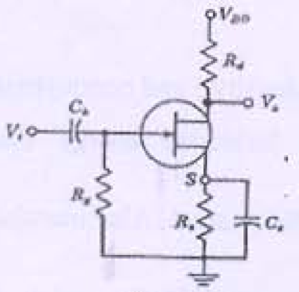

Figure-1

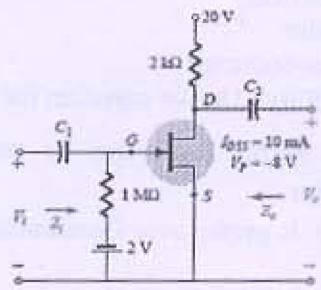

Figure-2

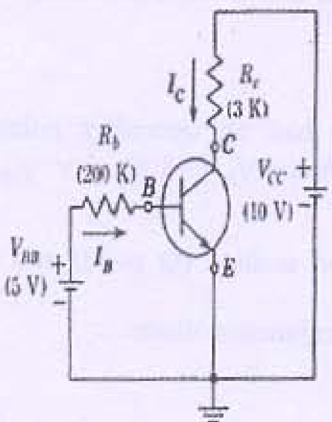

Figure-3

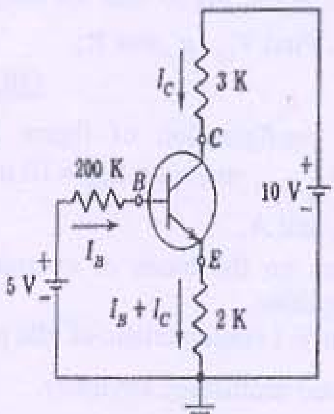

Figure-4

Parameter	CE	CC	CB
h_i	1 kΩ	1 kΩ	20 Ω
h_r	2.5×10^{-4}	≅1	3.0×10^{-4}
h_f	50	−50	−0.98
h_o	25 μA/V	25 μA/V	0.5 μA/V
$1/h_o$	40 kΩ	40 kΩ	2 MΩ

Table-1

******